Advanced Quantum Mechanics Assignment 3

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Problem 1

This is manifestly a three-degree-of-freedom system, thus, we start by writing the dynamical laws using the Newton's second law of motion:

$$m\partial_t^2 x_1 = -kx_1 + k(x_2 - x_1) \tag{1}$$
$$m\partial_t^2 x_2 = -k(x_2 - x_1) + k(x_3 - x_2) \tag{2}$$
$$m\partial_t^2 x_3 = -k(x_3 - x_2) - k(x_3) \tag{3}$$

Let us write this equations in terms ofn vectors and matrices, and also defining ($\alpha = \frac{k}{m}$).

$$\partial_t^2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2\alpha & \alpha & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & \alpha & -2\alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \tag{4}$$

Next, we start to find the basis, in which the matrix above is diagnoalized, to do so we must solve the matrix for its eigenvalues:

$$\det(\Omega - \lambda I) = 0 \tag{5}$$

This would result in the following polynomial equation:

$$-4\alpha^3 - 10\alpha^2\lambda - 6\alpha\lambda^2 - \lambda^3 = 0. \tag{6}$$

Solving this for λ we have:

$$\lambda_1 \to -2\alpha, \quad \lambda_2 \to -\alpha(2 - \sqrt{2}), \quad \lambda_3 \to -\alpha(2 + \sqrt{2}) \tag{7}$$

What this means is that there's a basis in which x'_1, x'_2, x'_3 follow this dynamical law:

$$\begin{cases} \partial_t^2 x'_1 = -2\alpha x'_1 \\ \partial_t^2 x'_2 = -\alpha(2 - \sqrt{2}) x'_2 \\ \partial_t^2 x'_3 = -\alpha(2 + \sqrt{2}) x'_3 \end{cases} . \tag{8}$$

Defining

$$\omega_1 := 2\alpha, \quad \omega_2 := \alpha(2 - \sqrt{2}), \quad \omega_3 := \alpha(2 + \sqrt{2}). \tag{9}$$

Thus we've got three de-coupled ordinary differential equations to solve:

$$\partial_t^2 x'_1 = -\omega_1 x'_1 \tag{10}$$
$$\partial_t^2 x'_2 = -\omega_2 x'_2 \tag{11}$$
$$\partial_t^2 x'_3 = -\omega_3 x'_3, \tag{12}$$

which have the solutions below:

$$x'_1 = A_1 \cos(\sqrt{\omega_1} t + \phi_1) \tag{13}$$
$$x'_2 = A_2 \cos(\sqrt{\omega_2} t + \phi_2) \tag{14}$$
$$x'_3 = A_3 \cos(\sqrt{\omega_3} t + \phi_3) \tag{15}$$

Now it's a good time to notice that x'_1, x'_2, and $x'3$ can be written in terms of our first variables x_1, x_2, x_3 since we know that they satisfy the relation:

$$\Omega x'_i = \lambda_i x'_i \tag{16}$$

And we have the matrix Ω in the original basis, we then have:

$$\begin{pmatrix} -2\alpha & \alpha & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & \alpha & -2\alpha \end{pmatrix} x'_i = \lambda_i x'_i \tag{17}$$

Solving for each (λ_i, x'_i) we find each vector in the x_1, x_2, x_3 basis as:

$$x'_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad x'_2 = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \quad x'_3 = \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \tag{18}$$

This can be written more nicely in Dirac notation:

$$|1'\rangle = -|1\rangle + |3\rangle \tag{19}$$

$$|2'\rangle = |1\rangle + \sqrt{2}\,|2\rangle + |3\rangle \tag{20}$$

$$|3'\rangle = |1\rangle - \sqrt{2}\,|2\rangle + |3\rangle \tag{21}$$

Using this we solve for $|1\rangle, |2\rangle, |3\rangle$, leading to:

$$|1\rangle = \frac{1}{4}\left(-2\,|1'\rangle + |3'\rangle + |2'\rangle\right) \tag{22}$$

$$|2\rangle = \frac{1}{4}\left(-\sqrt{2}\,|3'\rangle + \sqrt{2}\,|2'\rangle\right) \tag{23}$$

$$|3\rangle = \frac{1}{4}\left(2\,|1'\rangle + |3'\rangle + |2'\rangle\right) \tag{24}$$

Substituting and we have the solutions:

$$|1\rangle = \frac{1}{4}\left(-2A_1\cos\left(\sqrt{\frac{2k}{m}}t + \phi_1\right) + A_3\cos\left(\sqrt{\frac{\left(2+\sqrt{2}\right)k}{m}}t + \phi_3\right) + A_2\cos\left(\sqrt{\frac{\left(2-\sqrt{2}\right)k}{m}}t + \phi_2\right)\right) \tag{25}$$

$$|2\rangle = \frac{1}{4}\left(-\sqrt{2}A_3\cos\left(\sqrt{\frac{\left(2+\sqrt{2}\right)k}{m}}t + \phi_3\right) + \sqrt{2}A_2\cos\left(\sqrt{\frac{\left(2-\sqrt{2}\right)k}{m}}t + \phi_2\right)\right) \tag{26}$$

$$|3\rangle = \frac{1}{4}\left(2A_1\cos\left(\sqrt{\frac{2k}{m}}t + \phi_1\right) + A_3\cos\left(\sqrt{\frac{\left(2+\sqrt{2}\right)k}{m}}t + \phi_2\right) + A_2\cos\left(\sqrt{\frac{\left(2-\sqrt{2}\right)k}{m}}t + \phi_3\right)\right) \tag{27}$$

Given the 6 boundary conditions one can find the position of each mass at any given time. $\square$

Problem 2

We want to solve for the eigenvalues of the matrix below:

$$\Omega = \begin{pmatrix} -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} \end{pmatrix} \tag{28}$$

To do so we first calculate the following Dirac determinant:

$$\det\left(\Omega - \lambda I\right) \tag{29}$$

which is:

$$= \left(\frac{2k}{m} + \lambda\right)\left(\frac{2k}{m} + \lambda\right) - \frac{k^2}{m^2} \tag{30}$$

$$= \lambda^2 + \frac{4k}{m}\lambda + \frac{4k^2}{m^2} - \frac{k^2}{m^2} \tag{31}$$

$$= \lambda^2 + \frac{4k}{m}\lambda + \frac{3k^2}{m^2} \tag{32}$$

setting this to zero, and solving for lambda:

$$0 = \lambda^2 + \frac{4k}{m}\lambda + \frac{3k^2}{m^2} \tag{33}$$

$$\lambda = \begin{cases} -\frac{3k}{m} \\ -\frac{k}{m} \end{cases} \tag{34}$$

Now solving:

$$\Omega\,|I\rangle = \lambda_I\,|I\rangle \tag{35}$$

leads to:

$$\Omega\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} \tag{36}$$

$$\rightarrow \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} = -\frac{3k}{m}\begin{pmatrix} a \\ b \end{pmatrix} \tag{37}$$

and similarly for λ_{II} and $|II\rangle$:

$$\Omega\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} \tag{38}$$

$$\rightarrow \begin{pmatrix} -2\frac{ak}{m} + \frac{bk}{m} \\ \frac{ak}{m} - \frac{2bk}{m} \end{pmatrix} = -\frac{k}{m}\begin{pmatrix} a \\ b \end{pmatrix} \tag{39}$$

solving the first system of equations leads to:

$$|I\rangle = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{40}$$

and the second one:

$$|II\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{41}$$

Which if normalized would become:

$$|I\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \tag{42}$$

$$|II\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{43}$$

$$\square \tag{44}$$

Problem 3

One can write the two eigenstates as:

$$|I\rangle = \frac{1}{\sqrt{2}}\left(-\,|1\rangle + |2\rangle\right) \tag{45}$$

$$|II\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right) \tag{46}$$

where:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} * |2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{47}$$

Therefore, for the coupled spring problem since:

$$|I\rangle = A_I \cos\left(\sqrt{\frac{3k}{m}}t\right) \tag{48}$$

$$|II\rangle = A_{II} \cos\left(\sqrt{\frac{k}{m}}t\right) \tag{49}$$

We solve 45 and 46 for $|1\rangle, |2\rangle$ we get:

$$|1\rangle = \frac{1}{2}\left(-A_I \cos\left(\sqrt{\frac{3k}{m}}t\right) + A_{II}\cos\left(\sqrt{\frac{k}{m}}t\right)\right) \tag{50}$$

$$|2\rangle = \frac{1}{2}\left(A_I \cos\left(\sqrt{\frac{3k}{m}}t\right) + A_{II}\cos\left(\sqrt{\frac{k}{m}}t\right)\right) \tag{51}$$

$$\square \tag{52}$$

Problem 4

I couldn't see any "matrix" above.

Problem 5

We assume Ω is a Hermitian operator with $\|\Omega\| < 1$. Consider the partial sums

$$S_N = \sum_{n=0}^{N} \Omega^n. \tag{53}$$

Multiply by $(I - \Omega)$:

$$\begin{aligned}(I-\Omega)S_N &= (I-\Omega)\left(I + \Omega + \Omega^2 + \cdots + \Omega^N\right) && (54)\\ &= I - \Omega^{N+1}. && (55)\end{aligned}$$

Since $\|\Omega\| < 1$, we have

$$\|\Omega^{N+1}\| \le \|\Omega\|^{N+1} \longrightarrow 0 \qquad (N \to \infty). \tag{56}$$

Taking the limit of both sides,

$$(I-\Omega)\left(\sum_{n=0}^{\infty} \Omega^n\right) = I. \tag{57}$$

Hence the series converges and its sum is the inverse of $(I - \Omega)$:

$$\sum_{n=0}^{\infty} \Omega^n = (I-\Omega)^{-1}. \tag{58}$$

Thus the geometric series formula holds for Hermitian operators whose norm is strictly less than 1.

Problem 6

Recall that H is Hermitian if $H^\dagger = H$, and U is unitary if $U^\dagger U = I$. We need to show that $(e^{iH})^\dagger e^{iH} = I$. Therefore, first we calculate the adjoint of e^{iH}:

$$\begin{aligned}(e^{iH})^\dagger &= \left(\sum_{n=0}^{\infty} \frac{(iH)^n}{n!}\right)^\dagger && (59)\\ &= \sum_{n=0}^{\infty} \frac{((iH)^n)^\dagger}{n!} && (60)\\ &= \sum_{n=0}^{\infty} \frac{(i^*)^n (H^\dagger)^n}{n!} && (61)\\ &= \sum_{n=0}^{\infty} \frac{(-i)^n H^n}{n!} && (62)\\ &= e^{-iH} && (63)\end{aligned}$$

Now the product:

$$
\begin{aligned}
(e^{iH})^\dagger e^{iH} &= e^{-iH} e^{iH} && (64)\\
&= e^{-iH+iH} && (65)\\
&= e^{0} && (66)\\
&= I && (67)
\end{aligned}
$$

a. Therefore, e^{iH} is unitary. $\square$

Problem 7

Problem 8: Sakurai 7

Matrix elements of $|\alpha\rangle\langle\beta|$ are:

$$|\alpha\rangle\langle\beta|_{ij} = \langle a_i|\alpha\rangle\langle\beta|a_j\rangle \tag{68}$$

the latter braket can be written in terms of the conjugate:

$$\langle a_i|\alpha\rangle\langle\beta|a_j\rangle \quad \square \tag{69}$$

b. For the sping half in the $|\pm\rangle_z$ we've got:

$$\langle+|S_z\rangle = 1, \quad \langle-|S_z\rangle = 0, \quad \langle\pm|S_x\rangle = \tfrac{1}{\sqrt{2}} \tag{70}$$

therefore:

$$|S_z\rangle\langle S_x| = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} \tag{71}$$

$\square$.

Problem 9: Sakurai 8

Assuming:

$$A\,(|i\rangle + |j\rangle) = \alpha_i\,|i\rangle + \alpha_j\,|j\rangle \tag{72}$$

the latter ket:

$$|x\rangle = \alpha_i\,|i\rangle + \alpha_j\,|j\rangle \tag{73}$$

has a different direction with respect to $|i\rangle + |j\rangle$ therefore, it cannot be an eigenstate (eigenvector) for A unless,

$$\alpha_j = \alpha_i \tag{74}$$

It also can't be a third eigenstate because otherwise the orthogonality of these eigenstates would be of question.

Problem 10: Sakurai 9

a. For any state $|a\rangle$:

$$\prod_{a'}(A - a')\,|a\rangle = \prod_{a'}(A - a')\sum_i |a_i\rangle\langle a_i|a\rangle \tag{75}$$

which can be written as:

$$\sum_i\prod_{a'}(a_i - a')\,|a_i\rangle\langle a_i|a\rangle = \sum_i 0 = 0 \tag{76}$$

If the product instead is over all $a_j - a'$ then the only surviving term in the sum is:

$$\prod_{a'}(a_j - a')\,|a_i\rangle\langle a_i|a\rangle \tag{77}$$

and dividing by the factors $a_j - a'$ just gives the projection of $|a\rangle$ on the direction $|a'\rangle$. For the operator $A := S_z$ and $\{|a'\rangle\}$, we have:

$$\prod_{a'}(A - a') = \left(S_z - \frac{\hbar}{2}\right)\left(S_z + \frac{\hbar}{2}\right) \tag{78}$$

Thus:

$$\prod_{a' \neq a''} \frac{A - a'}{a'' - a'} \begin{cases} = \frac{S_z + \frac{\hbar}{2}}{\hbar} \text{ for } a'' = +\frac{\hbar}{2} \\ = \frac{S_z - \frac{\hbar}{2}}{-\hbar} \text{ for } a'' = -\frac{\hbar}{2} \end{cases} \tag{79}$$

Problem 10: Sakurai 10

Using the orthonormality $\langle\pm|\pm\rangle = 1,\ \langle +|-\rangle = 0$we compute one representative commutator, e.g. $[S_x, S_y]$.

First form the product $S_x S_y$:

$$\begin{aligned} &S_x S_y &(80)\\ &= \frac{\hbar}{2}\frac{i\hbar}{2}\big(|+\rangle\langle -| + |-\rangle\langle +|\big)\big(-|+\rangle\langle -| + |-\rangle\langle +|\big) &(81)\\ &= \frac{i\hbar^2}{4}\big(-|+\rangle\langle -|+\rangle\langle -| + |+\rangle\langle -|-\rangle\langle +| - |-\rangle\langle +|+\rangle\langle -| + |-\rangle\langle +|-\rangle\langle +|\big). &(82) \end{aligned}$$

By orthonormality all cross terms $\langle\pm|\mp\rangle$ vanish and $\langle\pm|\pm\rangle = 1$, so

$$S_x S_y = \frac{i\hbar^2}{4}\big(-|-\rangle\langle +| - |+\rangle\langle -|\big) = -\frac{i\hbar^2}{4}\big(|+\rangle\langle -| + |-\rangle\langle +|\big). \tag{83}$$

Similarly

$$\begin{aligned} &S_y S_x &(84)\\ & &(85)\\ &= \frac{i\hbar}{2}\frac{\hbar}{2}\big(-|+\rangle\langle -| + |-\rangle\langle +|\big)\big(|+\rangle\langle -| + |-\rangle\langle +|\big) &(86)\\ &= \frac{i\hbar^2}{4}\Big(-|+\rangle\langle -|+\rangle\langle -| - |+\rangle\langle -|-\rangle\langle +| + |-\rangle\langle +|+\rangle\langle -| + |-\rangle\langle +|-\rangle\langle +|\Big)[4pt] &(87)\\ &= \frac{i\hbar^2}{4}\big(|-\rangle\langle +| + |+\rangle\langle -|\big). &(88) \end{aligned}$$

Now the commutator

$$\begin{aligned} [S_x, S_y] &= S_x S_y - S_y S_x &(89)\\ &= -\frac{i\hbar^2}{4}\big(|+\rangle\langle -| + |-\rangle\langle +|\big) - \frac{i\hbar^2}{4}\big(|+\rangle\langle -| + |-\rangle\langle +|\big) &(90)\\ &= -\frac{i\hbar^2}{2}\big(|+\rangle\langle -| + |-\rangle\langle +|\big). &(91) \end{aligned}$$

Recognize $|+\rangle\langle -| + |-\rangle\langle +| = \frac{2}{\hbar}S_x$ and compare with $S_z = \dfrac{\hbar}{2}(|+\rangle\langle +| - |-\rangle\langle -|)$. A more convenient route is to use the Pauli algebra identity (which follows from the same projector manipulations)

$$\sigma_i \sigma_j = \delta_{ij}, \mathbb{I} + i\varepsilon_{ijk}\sigma_k, \tag{92}$$

together with $S_i = \dfrac{\hbar}{2}\sigma_i$. From this we get

$$[S_i, S_j] = \frac{\hbar^2}{4}[\sigma_i, \sigma_j] = \frac{\hbar^2}{4}(2i\varepsilon_{ijk}\sigma_k) = i\varepsilon_{ijk}\hbar\Big(\frac{\hbar}{2}\sigma_k\Big) = i\varepsilon_{ijk}\hbar, S_k, \tag{93}$$

which is the desired commutation relation.

For the anticommutator use the same identity:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}, \mathbb{I}, \tag{94}$$

hence

$$\{S_i, S_j\} = \frac{\hbar^2}{4}\sigma_i, \sigma_j = \frac{\hbar^2}{4}(2\delta_{ij}\mathbb{I}) = \frac{\hbar^2}{2}, \delta_{ij}, \mathbb{I}. \tag{95}$$

(If one views the right-hand side as an operator on the two-dimensional spin space, it is exactly $\frac{\hbar^2}{2}\delta_{ij}$ times the identity.)

Thus we have proven

$$[S_i, S_j] = i\varepsilon_{ijk}\hbar, S_k, \qquad \{S_i, S_j\} = \frac{\hbar^2}{2}, \delta_{ij}, \mathbb{I}, \tag{96}$$

as required.